Mounish Reddy

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Career Overview

Passionate about designing scalable cloud-native solutions and intelligent systems that enhance real-world experiences. Proficient in developing AI-driven applications using AWS. Eager to explore innovations in cloud computing, Security and AI-based Systems. Seeking impactful roles in Cloud/AI domains where I can contribute to cutting-edge technological advancement.

Technical Skills

Programming Languages: Python
Cloud Platforms: Amazon Web Services (AWS)
AI/Frameworks: PyTorch, LangChain, LangGraph, LangFuse, MCP, RAG, Scikit-learn, CNN, RNN, BiLSTM, MLP, NLP, Pandas, HuggingFace
Web Development(fundamentals): HTML, CSS, JavaScript, React, Streamlit, Flask, FastAPI
Databases: MySQL, PostgreSQL, DynamoDB, ChromaDB
Tools: VS Code, Cursor IDE, Antigravity IDE, Google Collab, Git, GitHub, Postman, Docker, N8N, Nmap,
Other Skills: Critical Thinking, Patience, Emotional Stability, Team Collaboration, Adaptability, Team Support

Education

VIT-AP University, MTech Integrated Computer Science and Engineering (5 years) 2021 - 2026
- Achieved a CGPA of 8.42 in a highly competitive program.

Narayana Junior College, 11th and 12th 2019 - 2021
- Graduated with a Marks of 968/1000.

Ratnam High School, 10th 2018 - 2019
- Graduated with a CGPA of 9.8.

Work Experience

AI Engineer at NexusIQ Solutions, Hyderabad March 2026 – Present
- Developing Generative AI applications and agentic workflows using LangGraph, LangChain, Gemini, and open-source LLMs for enterprise automation and intelligent information retrieval.
- Designing Retrieval-Augmented Generation (RAG) pipelines involving document processing, embedding generation, vector databases, semantic search, and context-aware response generation.
- Building AI-powered assistants with tool calling, conversational memory, prompt engineering, and workflow orchestration to support business and knowledge management applications.
- Integrating REST APIs, databases, and cloud services while contributing to model evaluation, performance optimization, testing, and deployment of production-ready AI solutions.

Projects

Multi-Agent College Management Assistant 2026
- Built a Multi-Agent College Management Assistant leveraging LangGraph, Streamlit, SQLite, ChromaDB, and Ollama (Qwen 2.5 8B) to automate academic information retrieval through natural language queries.
- Engineered an agent workflow comprising Guardrail, Supervisor, SQL Query, and Validator Agents for secure database access, SQL validation, policy-based authorization, and contextual response generation.
- Integrated Retrieval-Augmented Generation (RAG) using ChromaDB and BGE embeddings to answer institution-specific queries from college handbooks, regulations, and policy documents.
- Implemented role-based access control, conversational memory, and multi-step reasoning workflows to improve response accuracy and user experience across student and faculty interactions.

Enterprise Financial Report Intelligence Platform (RAG) 2026
- Developed a production-grade Retrieval-Augmented Generation (RAG) platform using Flask, React, ChromaDB, and Gemini APIs to enable natural language querying of 350+ page enterprise financial reports.
- Implemented an end-to-end retrieval pipeline including PDF ingestion, intelligent chunking, BGE embedding generation, vector indexing in ChromaDB, semantic search, and context-aware response generation with source citations.

- Engineered advanced document intelligence features such as Retrieval Inspector, Prompt Explorer, similarity-based ranking, page-level source attribution, and conversational memory for accurate and explainable AI responses.
- Designed an enterprise-grade frontend with React, TailwindCSS, Framer Motion, and Three.js, featuring interactive analytics dashboards, retrieval visualizations, and modern AI-powered user experiences.

Heart Disease Risk Classification 2026

- Developed an end-to-end Machine Learning web application for heart disease risk classification using an optimized ensemble model combining PyTorch MLP, TabPFN, and Particle Swarm Optimization (PSO).
- Achieved 98 percent prediction accuracy through feature optimization, ensemble learning, hyperparameter tuning, and comprehensive model evaluation.
- Built a modern Flask-based dashboard enabling real-time medical risk assessment, prediction explanations, and patient-centric analytics.
- Integrated REST APIs, interactive visualizations, and responsive glassmorphism-based UI/UX components for seamless healthcare decision support.

Deepfake Face Detection System 2025

- Developed a full-stack AI web platform with user authentication, dashboards, model switching, and real-time image/video analysis using CNN-RNN architecture (ResNeXt50 + BiLSTM + Attention), achieving 95 percent detection accuracy.
- Trained and evaluated models on Celeb-DF, YouTube-Real, and FaceForensics++ datasets to improve robustness across diverse deepfake generation techniques.
- Implemented confidence scoring, visual analytics, and explainability features to assist users in understanding model predictions and detection outcomes.
- Built scalable backend APIs and interactive frontend dashboards for real-time inference, monitoring, and performance visualization.

AI File Translation using AWS Services 2024

- Designed a serverless file translation application pipeline using AWS Translate, Lambda, API Gateway, DynamoDB, Textract, and S3, and reducing manual effort by 80 percent.
- Implemented automated document ingestion, language detection, OCR extraction, and multilingual translation workflows for processing diverse file formats.
- Improved security with AWS IAM, KMS encryption, and CloudTrail auditing, reducing overall security risks by 40 percent.
- Deployed a scalable event-driven architecture with monitoring, logging, and fault-tolerant processing to support enterprise-grade translation workloads.

Achievements

- Member of the Bulls & Bears Club, participated in discussions on equity markets, financial analysis, investment strategies, and market research activities.
- Published research paper titled *"Heart Disease Risk Prediction Using TabPFN, MLP and PSO Based Ensemble"* in the International Journal of Engineering Research and Technology (IJERT).

Certifications

- Ultimate AWS Solutions Architect Associate(SAA-C03) – Udemy
- Ultimate AWS AI Practitioner(AIF-C01) – Udemy
- N8n: No Code AI Agent Builder - Simplilearn
- DevOps with AWS: Tools for Automated Workflows – LinkedIn
- Essentials of prompt Engineering - AWS skillbuild
- Fullstack Development using AI Workshop - Cuvette